

MK110 Plus 02

Development Document

Version V1.0

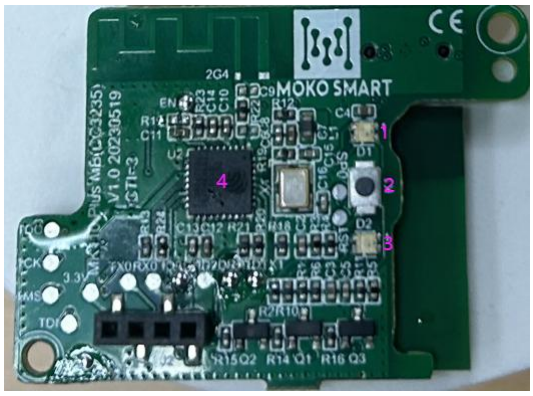
This document is provided for customers to develop their own firmware and flash the custom firmware on MK110 plus 02 hardware.

1. Hardware Description

The product MK110 plus 02 has two board: the power board, the control board. Firmware download can be done using only the control board.

Top View

control board

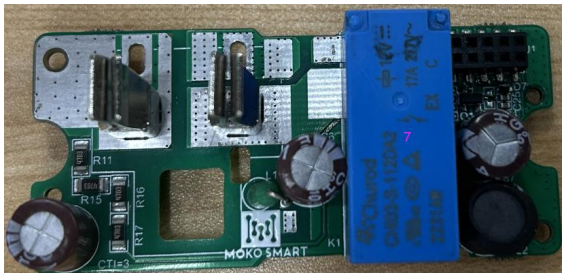


Bottom View

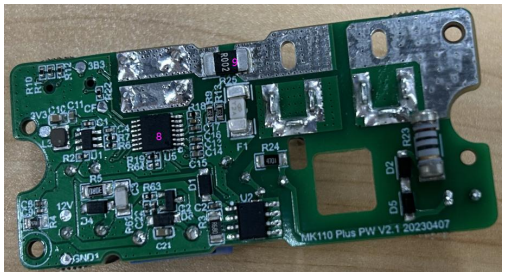
control board



power board



power board



Assembled

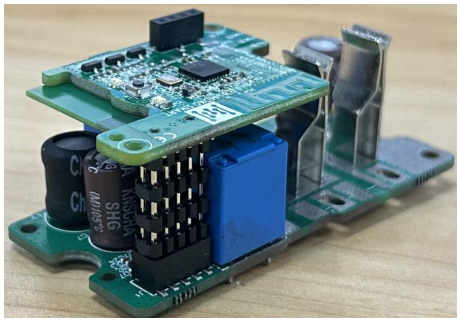


Figure 1 Hardware Photos of MK110 plus 02

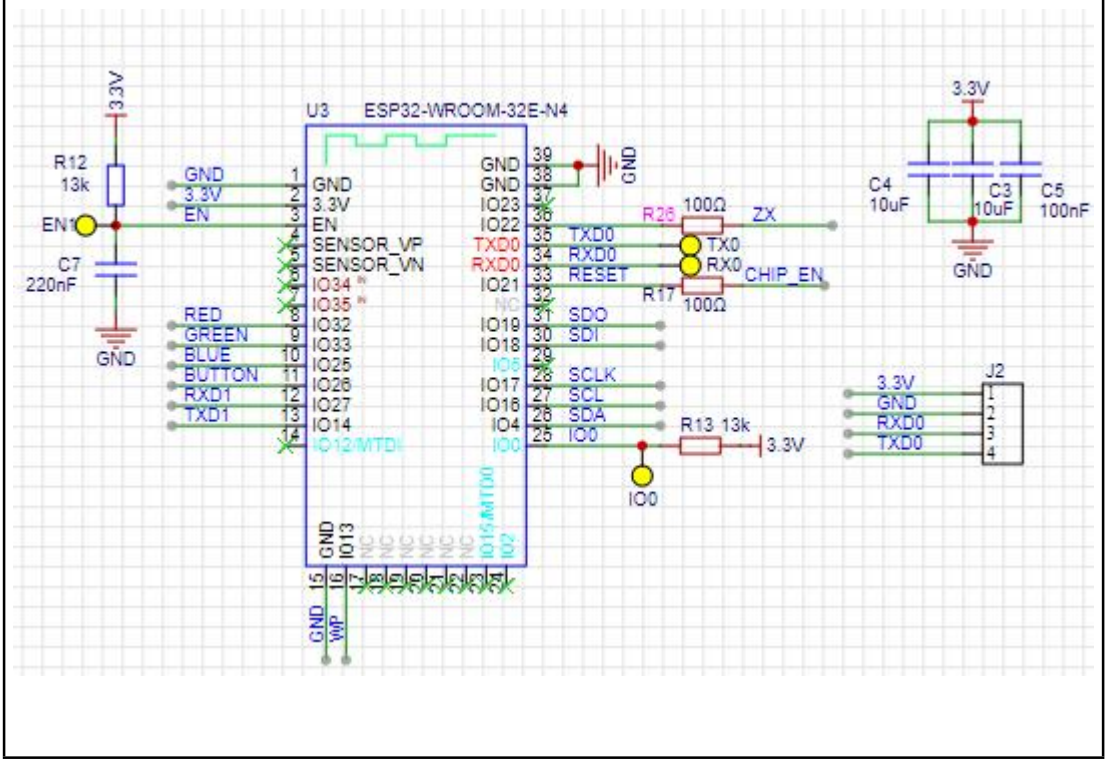
No.	Location	Name	Description
1	D1	LED1	1615RGB

2	K1(control board)	BUTTON	/
3	D2	LED2	1615RGB
4	U4	Slave chip	ESP32-C3FN4
5	U1	EEPROM	BL24C64A
6	U3	Main module	Wifi module(ESP32-WROOM-32E)
7	K1(power board)	RELAY	Normal open relay
8	U5	Single-phase Electrical Energy Meter IC	BL0942
9	R25	Resistor for current measure	2mR

2. Schematic Diagram of the Control Board

The WIFI module is ESP32-WROOM-32E,which is a ESPRESSIF module based on ESP32 SoC, The Slave module is ESP32-C3-MINI, which is a ESPRESSIF module based on ESP32-C3FH4 SoC. The schematic and pins definition are as below.

The Main module



The Slave module

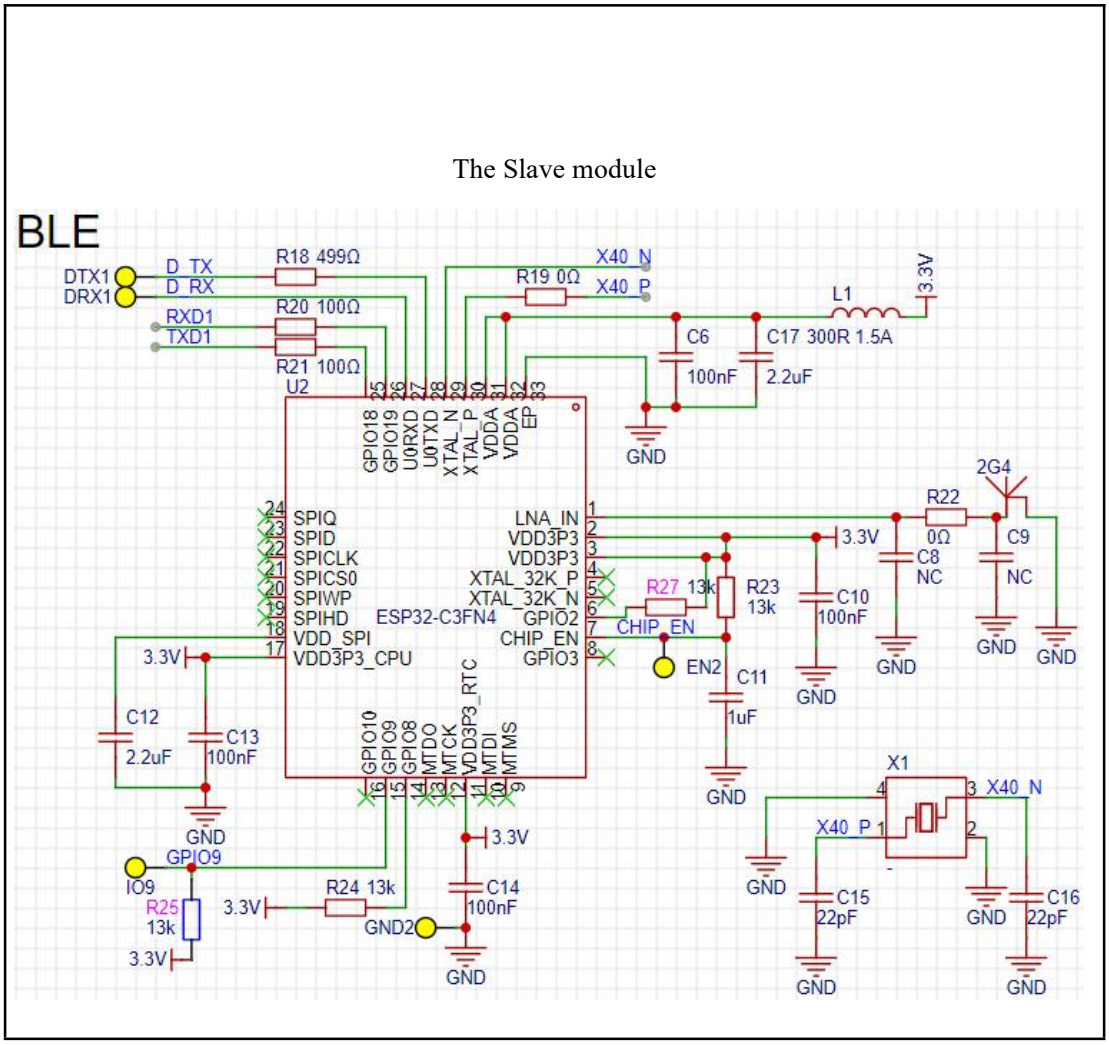


Figure 2 Schematic diagram of the Control module

The Main module interface definition

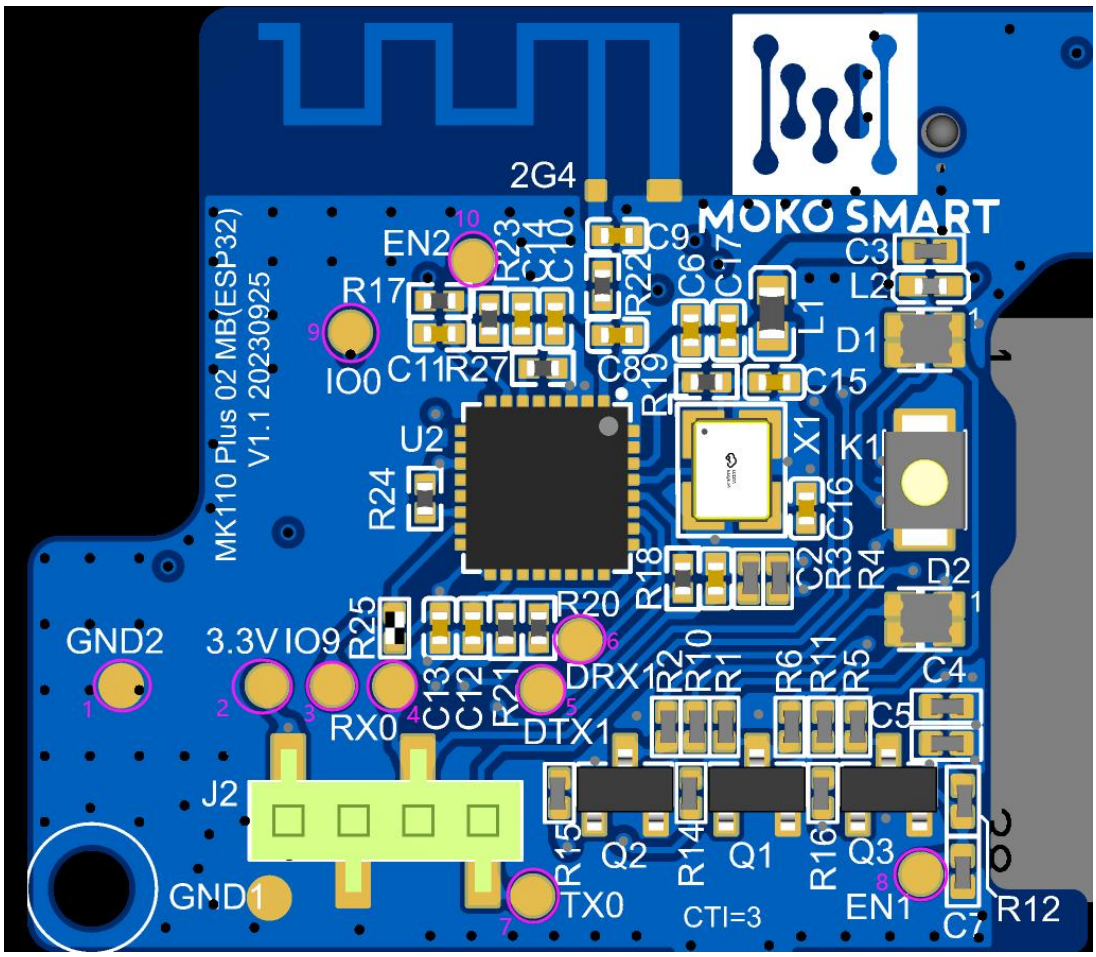
Pin no of ESP32	Pin of ESP32	Function	Notes
1&15&38 &39	GND	GND	GND
2	3.3V	3.3V	3.3V
3	EN	EN	reset pin. has reserved RC externally
8	IO32	RED	This pin is high, red on D1 and D2
9	IO33	GREEN	This pin is high, green on D1 and D2
10	IO25	BLUE	This pin is high, blue on D1 and D2
11	IO26	BUTTON	It's pulled up to 3.3V. When press down the button, the input signal on the pin is low level. It is labeled with K1.
12	IO27	RXD1	They are used for UART communication with

13	IO14	TXD1	ESP32-C3-MINI
16	IO13	WP	Write protect for BL24C64A.
26	IO4	SDA	The EEPROM is BL24C64A and supplied by 3.3V. It is labeled with U1.
27	IO16	SCL	
25	IO0	IO0	Please see figure 3 to get the wiring method
28	IO17	SCLK	They are used for SPI communication with BL0942
30	IO18	SDI	
31	IO19	SDO	
33	IO21	RESET	Control slave module reset
34	RXD0	RXD0	RXD0 is used to download and debug. Please see figure 3 to get the wiring method.
35	TXD0	TXD0	TXD0 is used to download and debug. Please see figure 3 to get the wiring method.
36	IO22	ZX	There is a normal-open relay on the hardware. If the pin output high level, the normal-open relay will be closed.

The Slave module interface definition

Pin no of ESP32-C3-MINI	Pin of ESP32-C3-MINI	Function	Notes
7	CHIP_EN	EN	Please see figure 3 to get the wiring method
15	GPIO9	GPIO9	
27	U0RXD	D_RX	
28	U0TXD	D_TX	
25	GPIO18	RXD1	They are used for UART communication with ESP32-WROOM-32E
26	GPIO19	TXD1	

3. Test Point Description



top side of the Control board

Figure 3 the test points on the control board

Test points in figure 3 are related to ESP32-WROOM-32E and ESP32-C3-MINI firmware download.

NO.	Test points on the board	Description
1	GND	Link to the GND
2	3.3V	Link to the 3.3V
9	IO0	Link to the GND of USB To Serial, Used to download firmware of the main module
7	TX0	Link to the RXD of USB To Serial, Used to download firmware of the main module

4	RX0	Link to the TXD of USB To Serial, Used to download firmware of the main module
3	IO9	Link to the GND of USB To Serial, Used to download firmware of the slave module
6	DRX	Link to the TXD of USB To Serial, Used to download firmware of the slave module
5	DTX	Link to the RXD of USB To Serial, Used to download firmware of the slave module
8	EN1	Link to the RTS of USB To Serial, Used to download firmware of the main module
10	EN2	Link to the RTS of USB To Serial, Used to download firmware of the slave module

4. Revision History

Revision	Description	Editor	Date
V1.0	Initial Version	TSX	2023.10.9

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